UF Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lecture 5B – Visual Sensory System II

EIN3241: Human Factors and Ergonomics
Spring 2026

February 23, 2026

Dr. Jad Atweh

Admin Issues

- Extra Session
  - Tomorrow!
  - 5:10 PM - 7:00 PM in Williamson 100
- Quiz 4
  - Friday
  - Lectures 5A, 5B, and 5C

Internal Factors Affecting Vision

- Internal factors relate to human physiological characteristics
  1. Dark adaptation
  2. Phorias
  3. Age
  4. Visual acuity
  5. Near point accommodation
  6. Peripheral vision
  7. Normal line of sight (NLS)
  8. Color discrimination
  9. Depth perception

2. Phorias

- Some people eyes converge too much, while others tend not to converge enough.
- Such conditions are called **phorias**.
- People with phorias also see one image but experience notable stress to the eye muscles, making them more vulnerable to rapidly developing visual fatigue.

3. Age

- With age, ciliary muscles weaken and visual performance declines.
- A young adult normally reads from a distance of about 30 cm (1 ft). At 60, the near point of accommodation is around 83cm (2.75 ft).
- In industry, reading requirements, including character size, may be based on the limiting age of the expected user population.

4. Visual Acuity

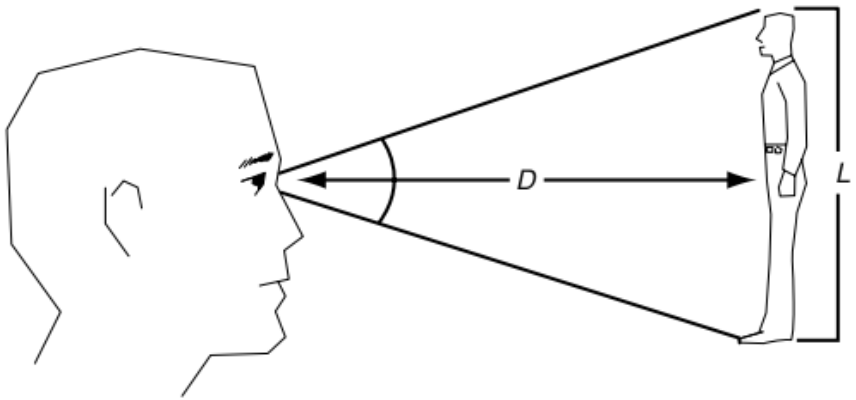
- Acuity is…
  - Ability to detect and discriminate fine details of objects at distance
  - The sharpness or clarity of vision.
  - Humans can discern 1 min. of visual arc

60 minutes = 1 degree

1 minute = $\frac{1}{60°}$

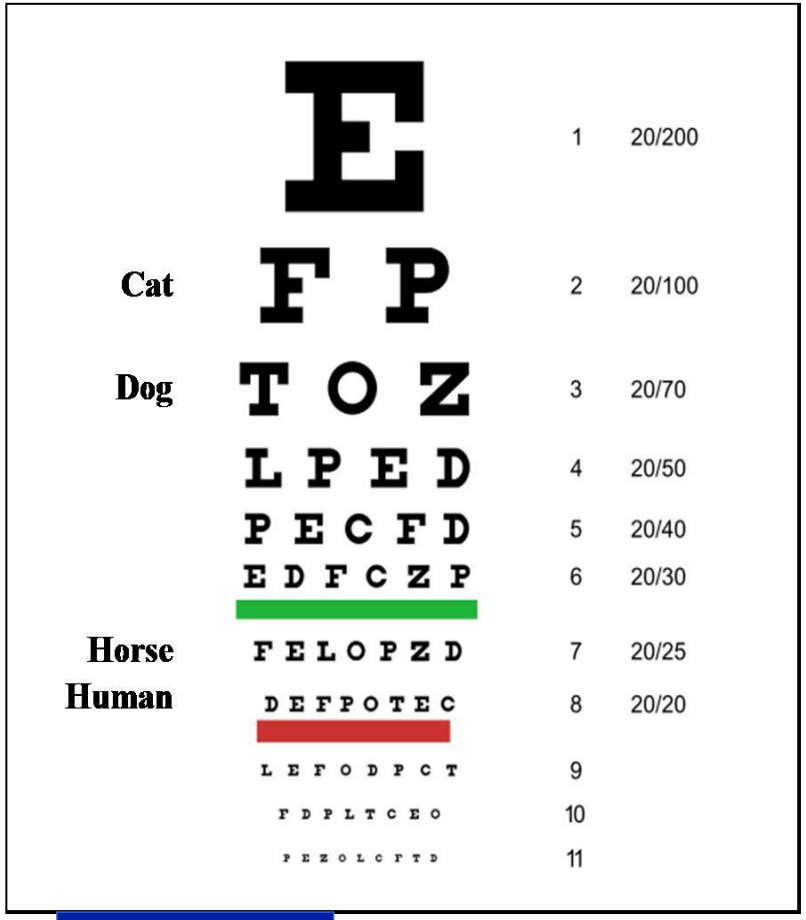
  - A **degree (°)** is divided into **60 minutes of arc (′)**
  - That means the smallest detail we can distinguish subtends an angle of 1/60 of a degree on the retina.
  - That's where the "20/20" standard comes from, being able to resolve 1′.

- Acuity changes with luminance and contrast
- How does visual acuity impact display size?
  - Degraded acuity requires larger characters for readability



4. Visual Acuity

- Snellen acuity is ratio of what you can see relative to what other humans can see (Common way to express visual acuity):
  - 20/20 vision:
    - Numerator is your distance from chart – typically held fixed at 20 ft
    - Denominator is distance at which person with normal eyesight could read smallest row in chart that you can read
- What does visual acuity 20/40 mean?
  - Line that you can correctly read at 20 ft can be read by "20/20" person at 40 ft
  - 20/40 = 0.5 ("half" of normal vision)
  - Equivalent to saying person possesses half spatial resolution capability as person with normal vision
    - Needs twice character size to discern type.



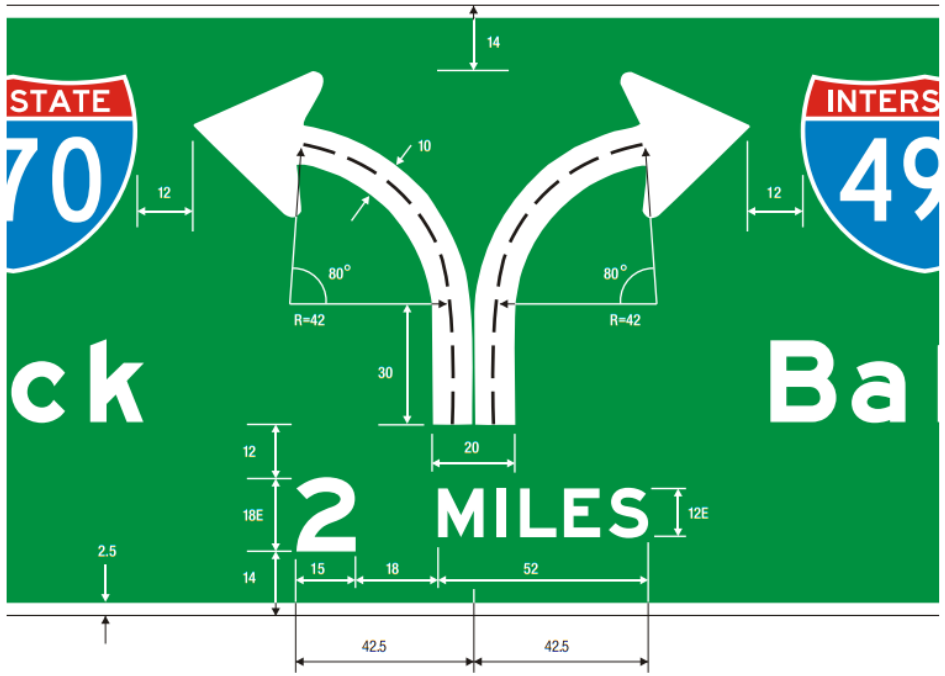
Snellen chart

Human Factors Application

- Highway signs are required to have certain font sizes
- DMV requires drivers to have acuity of 1.5 min. visual arc ("minimum separable acuity") or 20/30 vision
- Signs should have a legibility index of 30 ft/in

Graphic Component	Size
Lane Width	5" (125 mm) *
Lane Lines	1" x 6" (25 x 150 mm)
Space between Lane Lines	6" (150 mm)
Arrowheads	Standard "Up" Arrow
Stem Height (to upper point of depature—minimum)	30" (750 mm)
Space between arrowhead and route shield and between route shield and cardinal direction	12" (300 mm)*

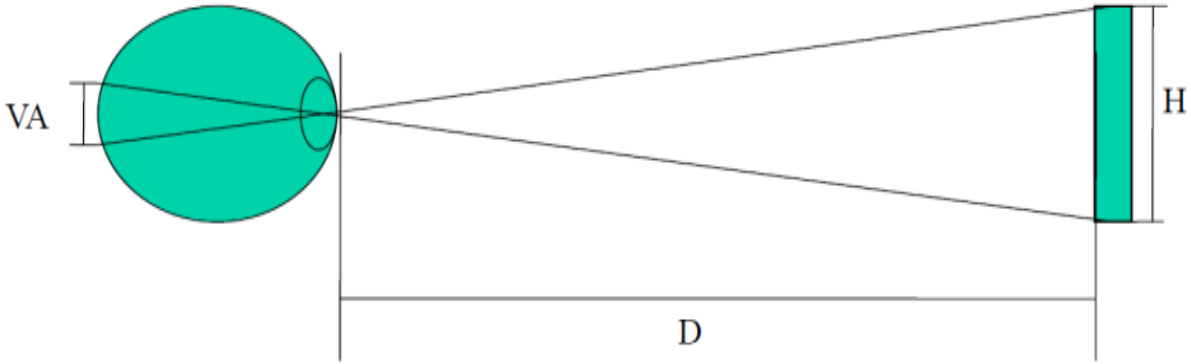
* Wider lane widths appear to better meet the needs of older drivers.



Visual angle

- Visual angle…
  - Angle that viewed object subtends at retina
  - Min discriminable visual angle = $\frac{1}{visual\ acuity}$
- Unit of measure:
  - Minutes of arc (due to shape of retina)

60 minutes = 1 degree
1 minute = $\frac{1}{60°}$

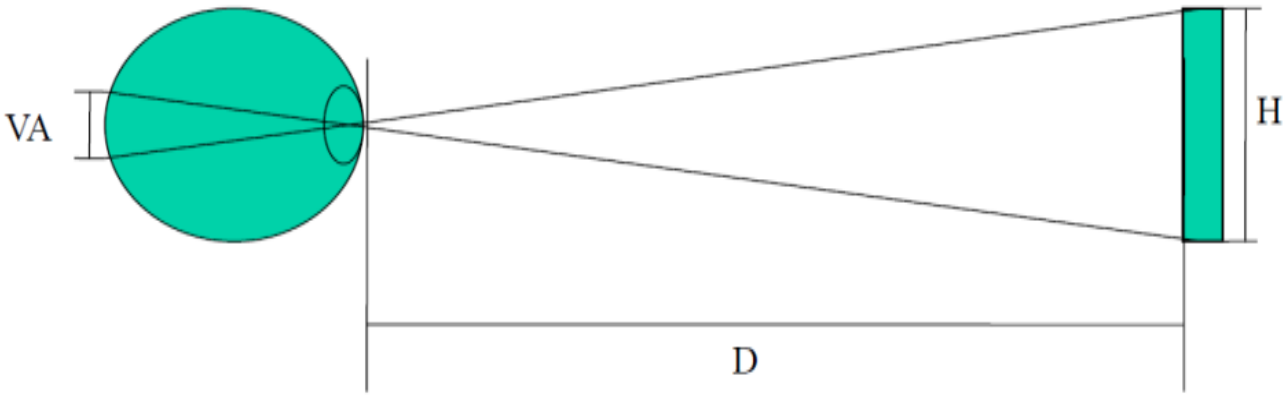


- Visual angle [degrees] = $2\ acrtan\left(\frac{H}{2D}\right)$
- For angles less than 10 degrees, an approximation is:
  - VA[min] = $\frac{3438 \times H}{D}$

Visual angle (cont.)

- Suppose we want an object to create a visual angle of 5 degrees on the retina, and the viewing distance is 20 inches.

1. Convert 5 degrees into minutes of arc: $5° \times 60 = 300$ *minutes*
2. Plug values into the equation: $300 = \frac{3438 \times H}{20}$
3. Solve for H (the object height): $H = \frac{300 \times 20}{3438} \approx 1.75$ *inches*



60 minutes = 1 degree

$$\text{VA[min]} = \frac{3438 \times H}{D}$$

Visual acuity example

- A worker's job is to detect a 0.01-inch space between two wires on an electronic component. One day the worker forgets his glasses (he has 20/50 uncorrected Snellen acuity). What is the farthest distance away the worker can be and still see the space between the wires?

- Worker visual acuity is 20/50 = 0.4 (below normal vision).
- Recall visual angle is reciprocal of acuity: Smallest visual angle he can resolve is 1/0.4 = 2.5 min. of arc.
- Visual Angle (min.) = (3438 * H) / D
- H = 0.01”
- Solve for D = 13.75”

Visual angle example

- What minimum visual angle can a person with 20/30 Snellen Acuity resolve?

a) 0.5 minute of arc

b) 1.0 minute of arc

c) 1.5 minute of arc

d) 2.0 minute of arc

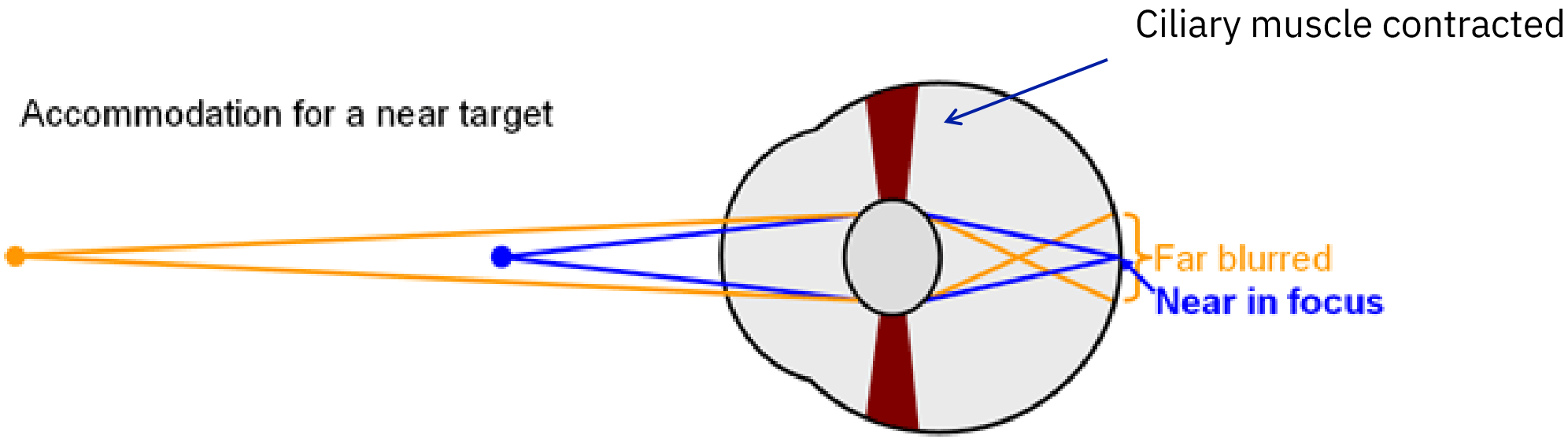
Acuity = 20/30 = 0.667

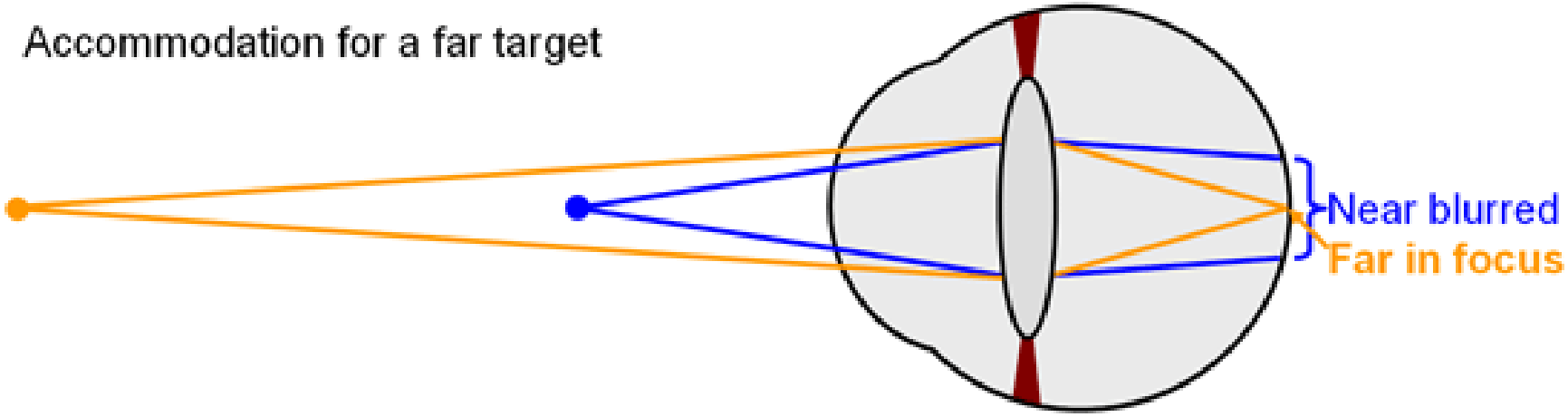
Angle = 1 / acuity = 1 / 0.667 = 1.5 arc min.

5. Near point of Accommodation

- Accommodation is the eye's ability to change the lens shape to focus on objects at different distances.
- Near point accommodation: The closest point an object can be brought in front of the eyes without losing details of the object (e.g., seeing it as two objects).
- Has been used as a measure of visual fatigue.
- Typical values
  - For a **young adult**, the near point is about **10 cm (4 inches)** from the eyes.
  - With **age**, the near point gets farther away (this is why older people need reading glasses)

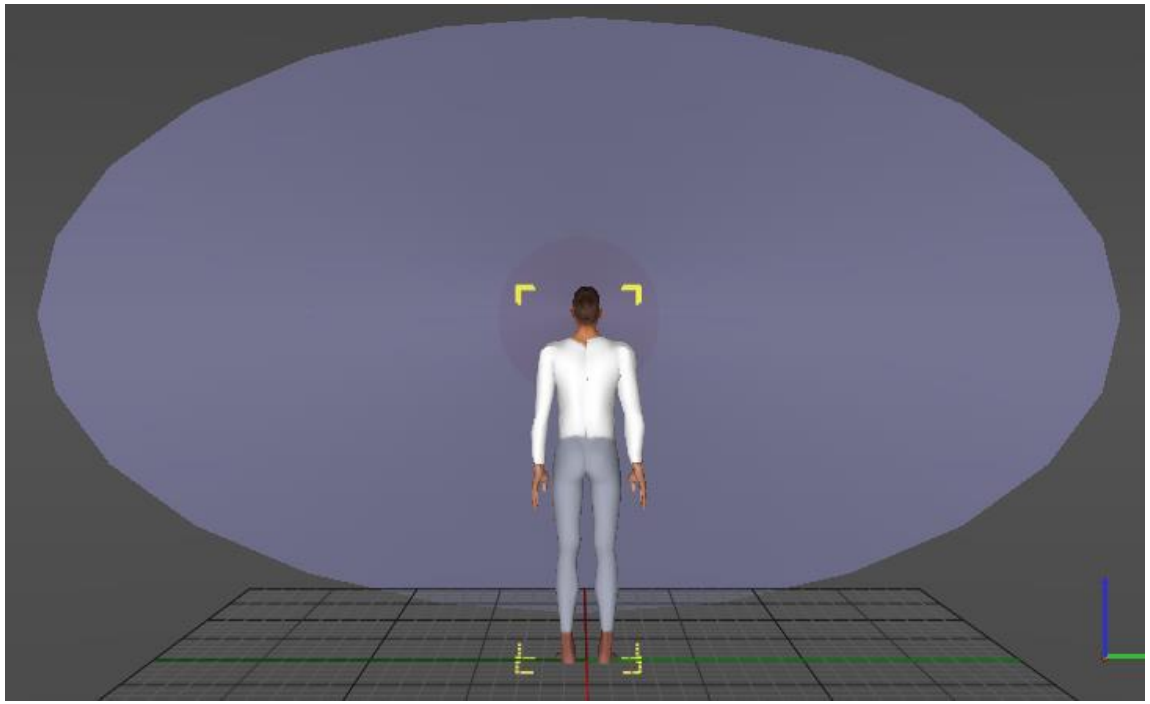
Accommodation

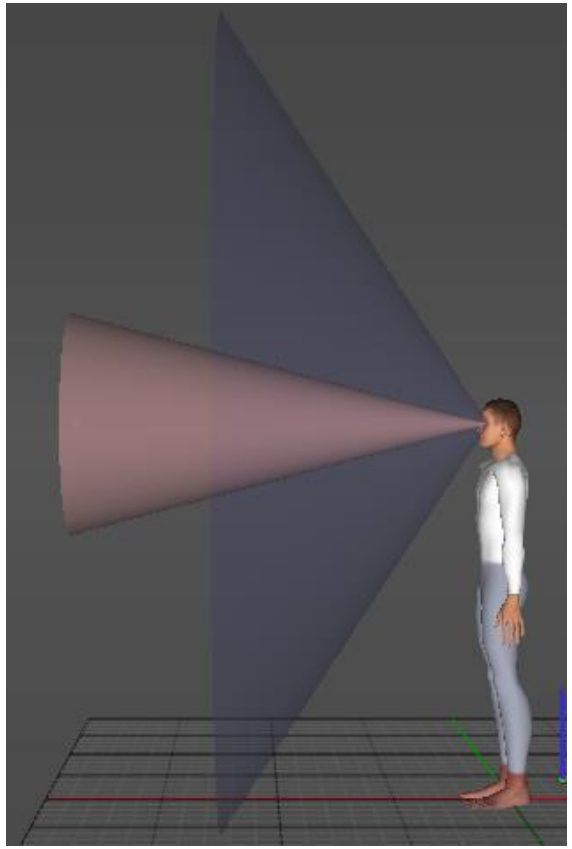




6. Peripheral Vision

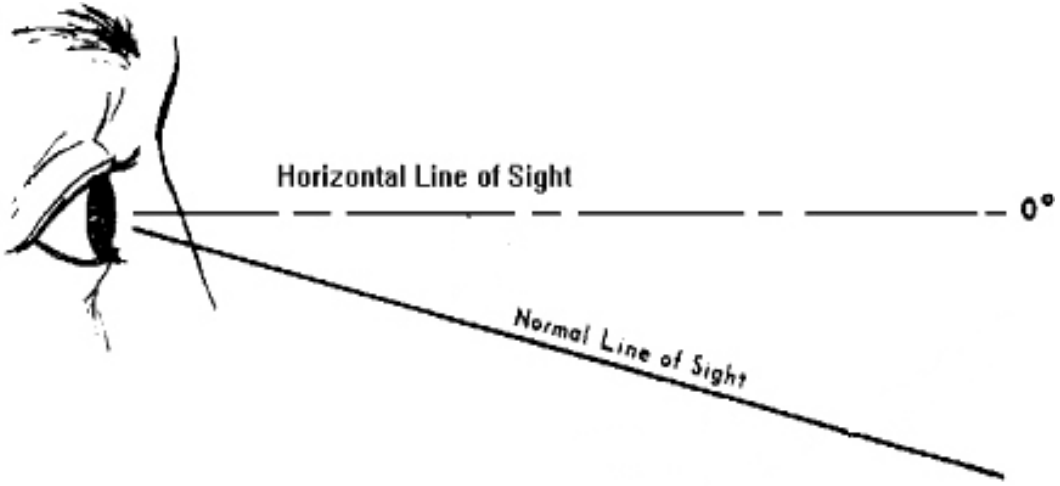
- With no eye or head movements, humans have a wider peripheral vision in the horizontal (208°) than in the vertical (160°).
- Assuming daylight conditions, acuity is sharpest at center.





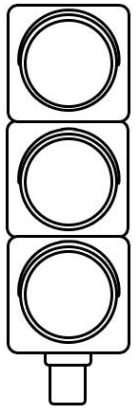
7. Normal Line of Sight (NLS)

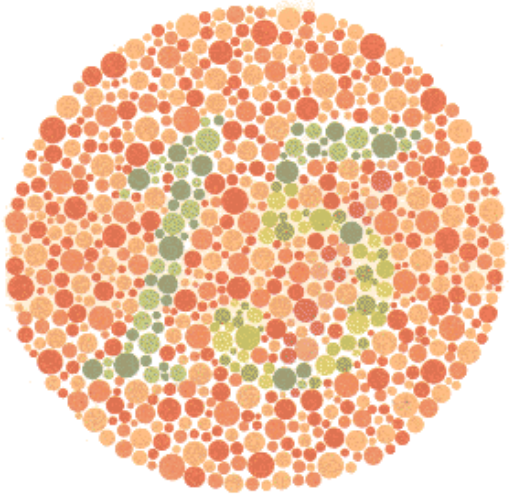
- When standing, NLS is 10°-15° below the horizontal.
- When seated, NLS is 20°-25° below the horizontal.



8. Color discrimination

- Cones in retina allow for perception of color
  - Three types of cones
  - Each type is sensitive to range of λ: (R (long), G (medium), B (short))
- People with normal color vision are trichromats
- People with color deficiency:
  - Dichromats (i.e., color blindness in one cone type)
  - Monochromat (i.e., color blindness in two cone types)
  - Males are more apt to color blindness (8% of the males vs 0.5% of the females!)
- If color blind, other visual cues may include shapes, letters…





https://www.color-blindness.com/coblis-color-blindness-simulator/

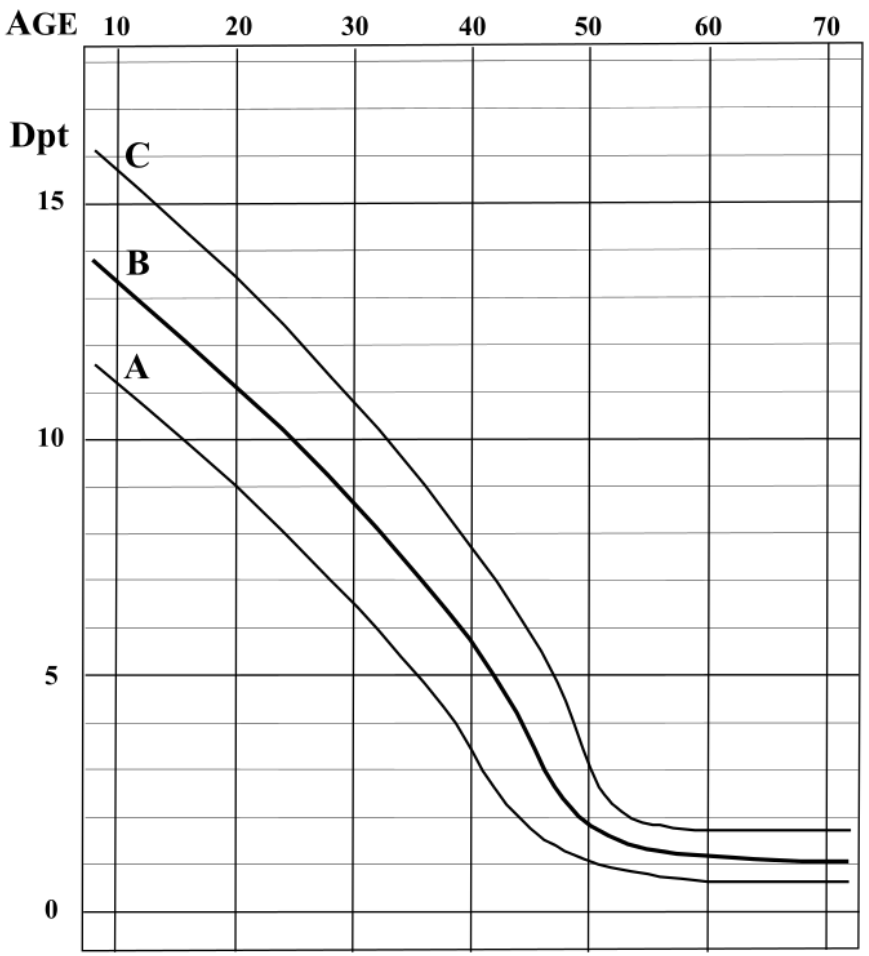
Perceptual processing: Bottom up / Top down

- Bottom up: Interpretation of stimulus driven by data in the world
- Top down: Interpretation of stimulus driven by knowledge in the head
- Bottom up: Yellow “slime” firetrucks can be easier to detect across multiple lighting conditions
- Top down: We expect firetrucks to be red which can speed detection and recognition of red firetrucks

9. Depth Perception: Accommodation

- Ability to focus objects at different distances through involuntary contraction/relaxation of ciliary muscles (reshaping lens)
- To focus on far objects, lens flattens, and muscles relaxes (light rays are nearly parallel; limited refraction for focus)
- To focus on near objects, lens become rounded and muscle contract (light rays diverge; greater refraction is needed for focus)
  - Can be fatiguing!

Accommodation Amplitude (Dpt) vs. Age

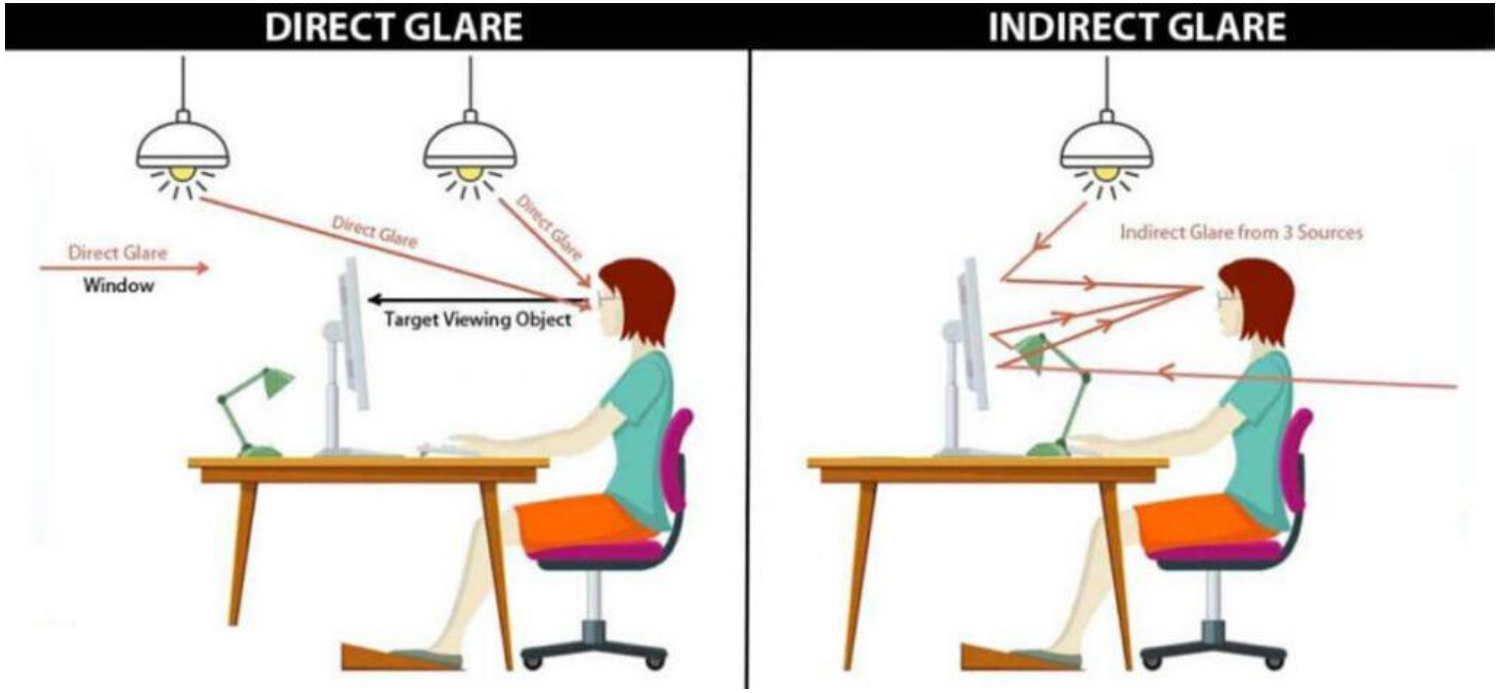


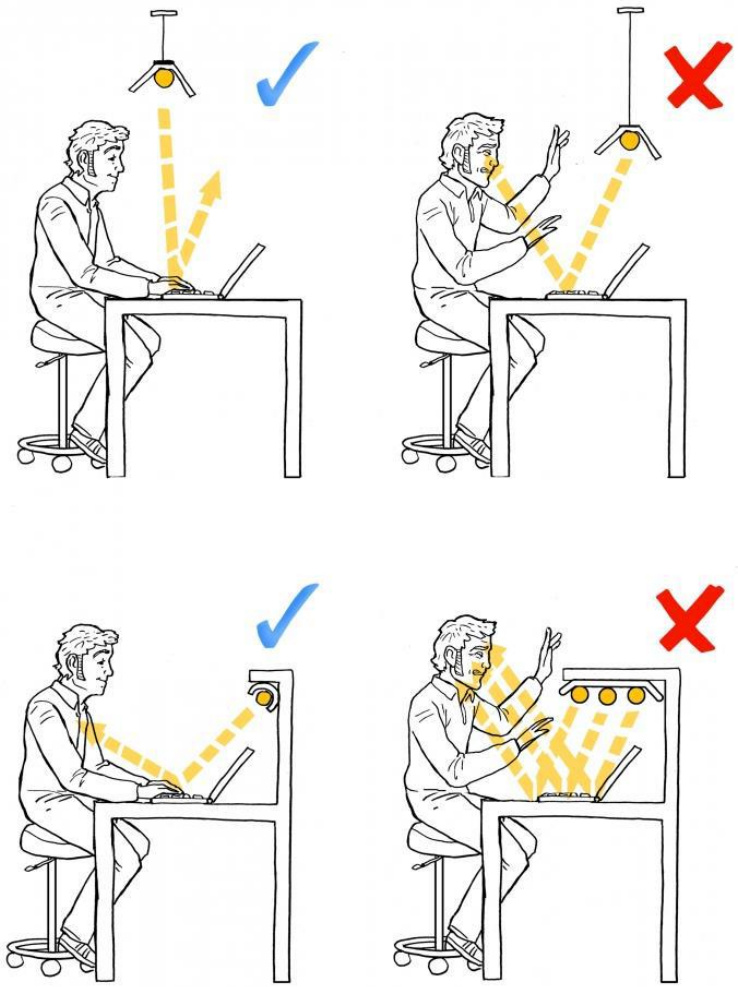
External Factors Affecting Vision

- External factors (we have more control of these factors):
  1. Position of target in the visual field
  2. Glare
  3. Color
  4. Color contrast
  5. Size of target
  6. Stability of target
  7. Amount of light
  8. Shift duration

Glare

- Proper workplace lighting reduces visual fatigue and errors.
- Glare can reduce contrast sensitivity, increase eye strain, and lower productivity.
- Glare can slow reaction times, obscure critical information (e.g., on screens or control panels).
- Older adults and individuals with visual impairments are more sensitive to glare. Addressing glare supports universal design principles and inclusivity in the workplace.





Contrast

- Contrast is perception of light and dark areas in image (luminance/color)
- Contrast is ratio between luminance (perception) of light (L) to dark (D) surfaces
  - Differences in luminance of perceived brightness of given object vs. background

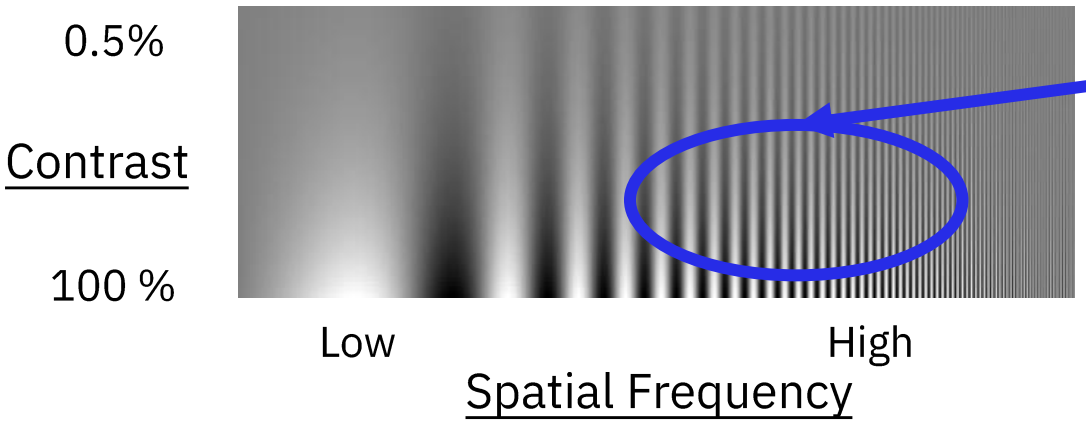
Lower contrast

Higher contrast

- Depends on spatial frequency – rate at which areas of stimuli alternate between light (white) and dark (black) per degree of visual angle [cycles per degree (CPD)].

Spatial frequency in contrast

- Spatial frequency describes how often light and dark (or color) patterns repeat across space.
- High Spatial Frequency = many cycles / degree of visual angle (many changes across space – detailed info)
- Low Spatial Frequency = few cycles / degree of visual angle (few changes across space – gross info)
- Spatial frequency grating is used to measure contrast sensitivity of humans (see below)

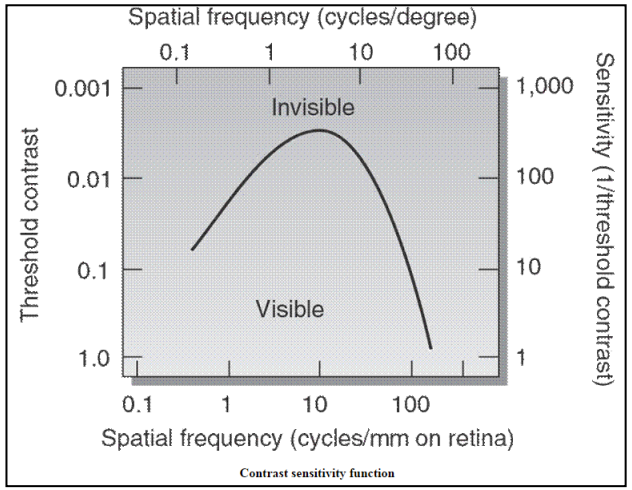


Our eyes are most sensitive to an intermediate range of spatial frequencies (around 2-4 CPDs) and less sensitive to spatial frequencies both lower and higher.

Contrast (cont.)

- Contrast sensitivity = reciprocal of minimum contrast (C_M) between lighter and darker area
  - C_M is contrast threshold = minimum amount of contrast detectable by observer
  - Critical in ability to detect target among background
  - Lower contrast threshold = higher sensitivity
  - Interacts with object focus in detection tasks

$$contrast\ sensitivity = \frac{1}{C_M}$$



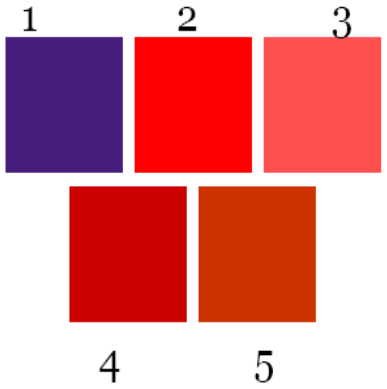
- Eyes are most sensitive to contrast at spatial freq. = 2-4 CPDs
- Eyes easily adapt (perceive differences in lightness) for frequencies in 2-6 CPDs (pupil adjusts to contrast)
- Problem: Frequency patterns (in sensitivity range) on surface of objects are primary stimulus for accommodation.
- Adaptation to spatial frequency gradient at surface of object leads to degraded accommodation for focus.

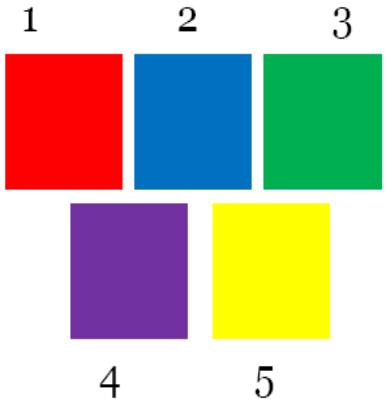
Color Contrast

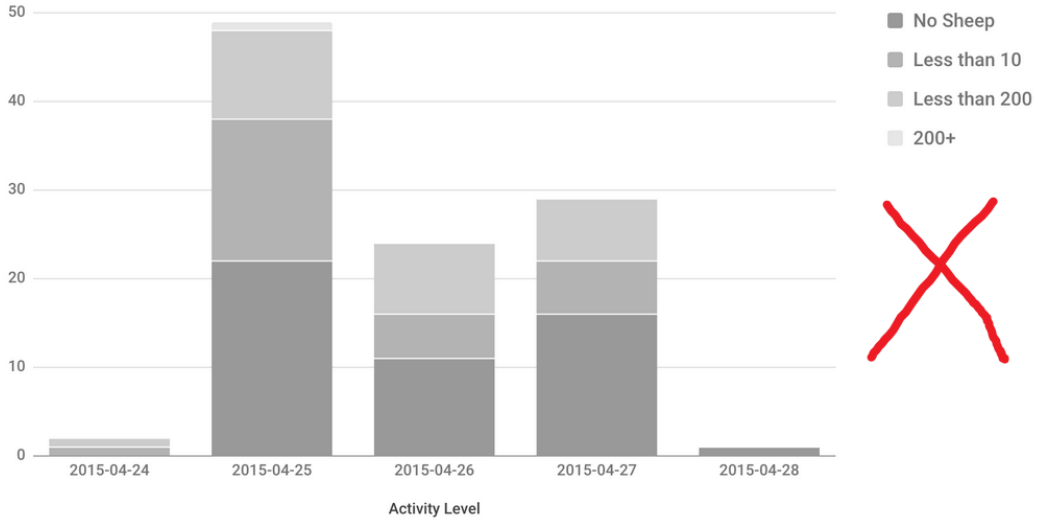
- Absolute: Is the square red?

- Difference: Are the squares alike or different?

- Choice: Pick the red square from each group

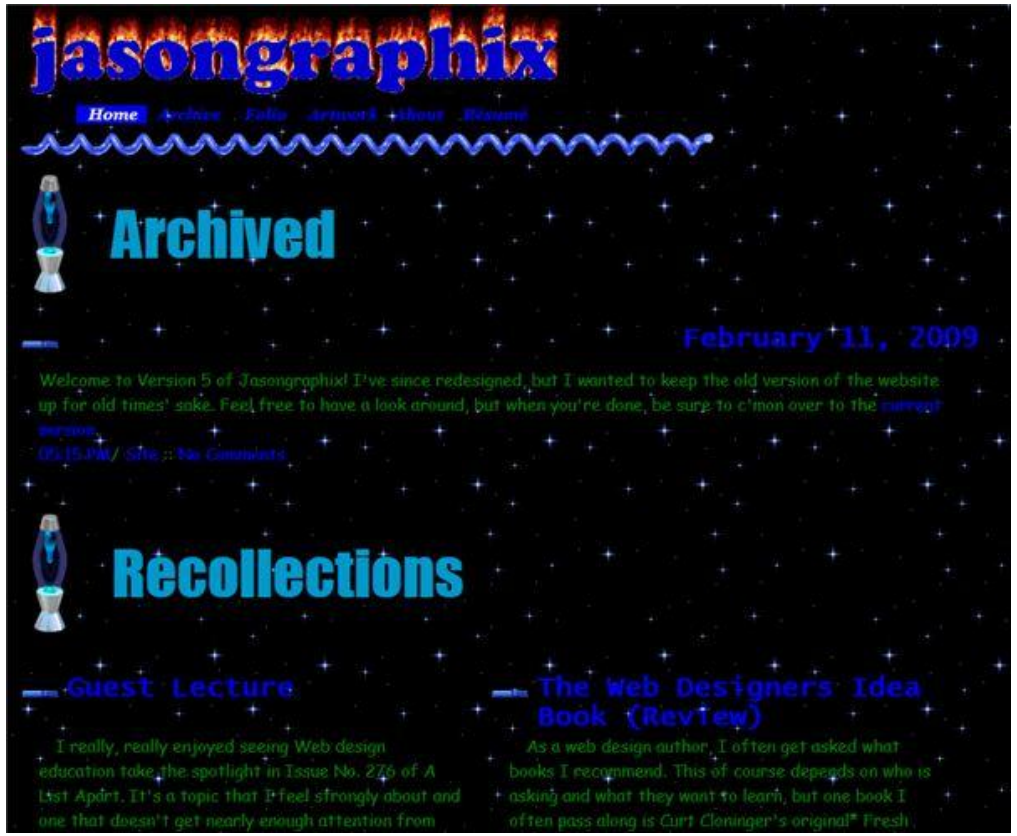






HF/E in screens and graphics

- Maximize color contrast
- Avoid saturated RGB stimuli on dark background
- Example of poor screen design…



Factors affecting contrast sensitivity and acuity

- Display features:
  - Luminance level (brightness)
  - Degree of contrast
- Task features:
  - Target object motion (dynamic visual acuity)
  - Exposure time
- Individual factors:
  - Age
  - Task training

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LEADING THE CHARGE, CHARGING AHEAD